

PART : CHEMISTRY

1. Arrange the following in ascending order wavelength

λ_1 = Infrared

λ_2 = Micro

λ_3 = X-ray

λ_4 = U.V.

(1) $\lambda_3 < \lambda_4 < \lambda_1 < \lambda_2$ (2) $\lambda_3 < \lambda_1 < \lambda_4 < \lambda_2$ (3) $\lambda_2 < \lambda_1 < \lambda_4 < \lambda_3$ (4) $\lambda_1 < \lambda_4 < \lambda_2 < \lambda_3$

Ans. (1)

Sol. Order of wavelength in EM spectrum:

Cosmic < Gamma < X-rays < UV < Visible < Infra Red < Micro < Radio

2. $t_{2g}^3 e_g^1$ configuration is possible in:

(1) WFL; high spin (2) WFL; low spin (3) SFL; high spin (4) SFL; low spin

Ans. (1)

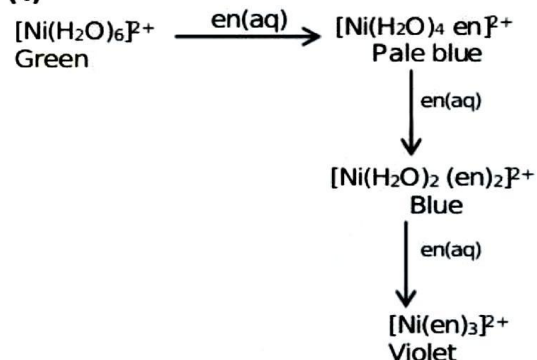
Sol. WFL will not cause pairing & above is high spin arrangement (greater no. unpaired e^-).

3. When ethane -1, 2-diammine is progressive added to aqueous of Nickel (II) chloride the sequence of colour changed observed will be

(1) Pale blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet (2) Violet $\rightarrow$ Blue $\rightarrow$ Pale blue $\rightarrow$ Green

(3) Pale blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green (4) Green $\rightarrow$ Pale blue $\rightarrow$ Blue $\rightarrow$ violet

Ans. (4)



Sol.

4. **Statement-I :** IE_1 of Sn $>$ Pb

Statement-II : IE_1 of Si $>$ Ge

(1) Both Statement I and statement II are true (2) Both statement I and statement II are false

(3) Statement I is true but statement II is false (4) Statement I is false but statement II is true

Ans. (4)

Sol. IE decreases down the gap. So Si $>$ Ge & Pb, Sn exceptional

5. Compound $\xrightarrow{\text{aqueous}} \text{B} \xrightarrow[\text{CH}_3\text{COOH}]{\text{KNO}_2} \text{Yellow ppt}$

(1) NiS

(2) ZnS

(3) CoS

(4) MnS

Ans. (3)

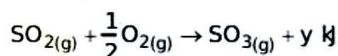
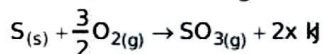
Sol. $\text{CoS} \xrightarrow{\text{aqueous}} \text{CoCl}_2 \xrightarrow[\text{CH}_3\text{COOH}]{\text{KNO}_2} \text{K}_3[\text{Co}(\text{NO}_2)_6] \downarrow \text{Yellow}$

6. 54.2% C, 9.2% H & 36.6% O are present in a compound. If its molar mass is 132 g, its molecular formula is
 (1) $C_6H_{12}O_3$ (2) $C_4H_8O_2$ (3) $C_6H_{12}O_6$ (4) None of these

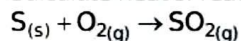
Ans. (1)

Sol. $C : H : O = \frac{54.2}{12} : \frac{9.2}{1} : \frac{36.6}{16}$
 $= 4.6 : 9.2 : 2.3$
 $= 2 : 4 : 1$
 $E_f = C_2H_4O_1 \quad (E.F.)_{mass} = 44$
 $n = \frac{132}{44} = 3 \therefore M_f = (C_2H_4O)_3 = C_6H_{12}O_3$

7. Consider the following reactions



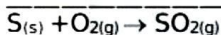
Calculate heat of reaction (kJ) for the given reaction



- (1) $-(x+y)$ (2) $-(2x+y)$ (3) x/y (4) $y - 2x$

Ans. (4)

Sol. $\Delta H_1 = -2x$
 $\Delta H_2 = -y$
 $S_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow SO_{3(g)}, \Delta H_1 = -2x$
 $SO_{3(g)} \rightarrow SO_{2(g)} + \frac{1}{2}O_{2(g)}, \Delta H_2 = y$



$$\Delta H_r = \Delta H_1 + \Delta H_2$$

$$= -2x + y$$

$$= y - 2x \text{ kJ}$$

8. Match the following cations with respective spin magnetic moment

Ions	H(B.M)
(i) Ti^{3+}	(p) 2.83
(ii) Sc^{+3}	(q) 0.00
(iii) V^{+2}	(r) 1.73
(iv) Ni^{+2}	(s) 3.87

(1) i-r; ii-q; iii-s; iv-p

(2) i-p; ii-q; iii-r; iv-s

(3) i-s; ii-p; iii-q; iv-r

(4) i-s; ii-p; iii-r; iv-q

Ans. (1)

9. Calculate the overall activation energy

$$K = \frac{k_1 k_3}{k_2} \quad E_{a_1} = 60 \text{ kJ}$$

$$E_{a_2} = 40 \text{ kJ}$$

$$E_{a_3} = 20 \text{ kJ}$$

Ans. (20)

Sol.

$$e^{-E_a/RT} = \sqrt{\frac{e^{-E_{a1}/RT} \cdot e^{-E_{a3}/RT}}{e^{-E_{a2}/RT}}}$$

$$e^{-E_a/RT} = \sqrt{e^{\frac{(E_{a2} - E_{a1} - E_{a3})}{RT}}}$$

$$-E_a = (E_{a2} - E_{a1} - E_{a3}) \times \frac{1}{2}$$

$$E_a = (E_{a1} + E_{a3} - E_{a2}) \times \frac{1}{2}$$

$$E_a = \frac{1}{2}(60 + 20 - 40) = 20 \text{ kJ}$$

- 10. Statement-I :** Oxygen-oxygen bond length in O_3 is greater than O_2 .
Statement-II : O—O bond order in O_3 is 1.5 and O—O bond order in O_2 is 2.
 (1) Both Statement I and statement II are true (2) Both statement I and statement II are false
 (3) Statement I is true but statement II is false (4) Statement I is false but statement II is true

Ans. (1)

- 11.** The successive ionisation energy (I.E.) of an element 'X' is given

	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅
X →	500	600	2000	2200	2600

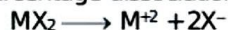
Data given in kJ/mol.

Find out the group number of element X.

- (1) Group-13 (2) Group-14 (3) Group-2 (4) Group-13

Ans. (3)

- 12.** MX_2 observed molar mass: 65.6 Normal molar mass: 164
 Find percentage dissociation.



Ans. (78)

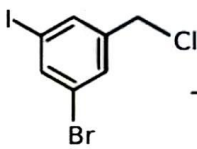
Sol. $i = \frac{\text{normal molar mass}}{\text{abnormal molar mass}} = \frac{164}{65.6} = 2.5$

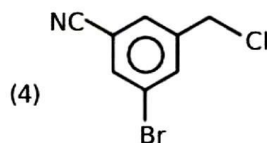
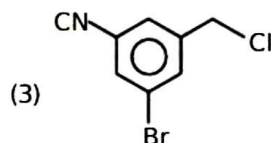
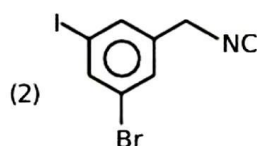
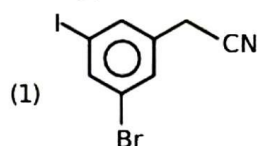
$$2.5 = 1 + 2\alpha$$

$$1.5 = 2\alpha$$

$$\alpha = 0.75$$

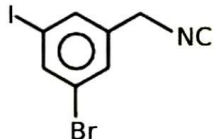
$$\% \alpha = 75$$

- 13.**  $\xrightarrow{\text{AgCN}}$ p Major Product?



Ans. (2)

Sol.



14. Match the following reactions given in Column-I with respective reagents given in Column-II.

Column-I

- (a) Etard Reaction
- (b) Gattermann Reaction
- (c) Gattermann Koch Reaction
- (d) Staphen Reaction

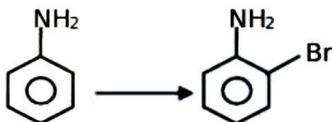
Column-II

- (p) $\text{SnCl}_2 + \text{HCl}$
- (q) CrO_2Cl_2
- (r) $\text{Cu} + \text{HCl}$
- (s) $\text{CO} + \text{HCl}$, Anhydrous AlCl_3

- (1) a-(q); (b)-(r); (c)-(s); (d)-(p)
- (2) a-(p); (b)-(q); (c)-(r); (d)-(s)
- (3) a-(q); (b)-(s); (c)-(p); (d)-(r)
- (4) a-(p); (b)-(r); (c)-(q); (d)-(s)

Ans. (1)

15.

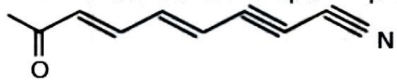


Above conversion can be done by using which reagents among the following.

- (1) Fe/Br_2 , $\text{H}_2\text{O}(\Delta)$, H_2SO_4
- (2) Ac_2O , H_2SO_4 , Br_2 , NaOH
- (3) Ac_2O , Br_2/AcOH , $\text{H}_2\text{O}/\text{H}^+$
- (4) Ac_2O , Br_2/Fe , NaOH

Ans. (2)

16. Find the total number of sp and sp^2 hybridised carbon atoms in the given compound.



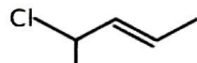
Ans. (8)

Sol. $\text{sp}^2 = 5$
 $\text{sp} = 3$
Total = 8

17. In the carius method, 0.25 gm organic compound is heated with fuming HNO_3 then AgNO_3 is added it gives 0.15 gm AgBr if molecular mass of AgBr is 188 then find mass percentage of Br in that organic compound

Ans. (25.53%)

18.



Find total number of stereoisomers of the given compound.

Ans. (4)

19. Match the following Nitrogenous Bases with their respective structures.

Column-I

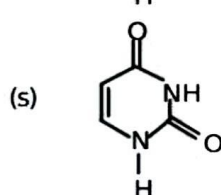
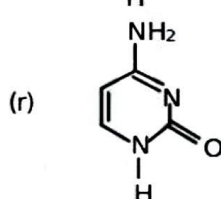
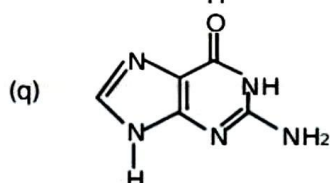
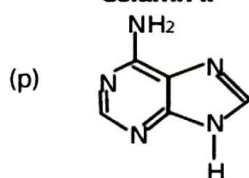
(a) Cytosine

(b) Uracil

(c) Guanine

(d) Adenine

Column-II



(1) a-(q); (b)-(r); (c)-(s); (d)-(p)

(2) a-(r); (b)-(s); (c)-(q); (d)-(p)

(3) a-(q); (b)-(s); (c)-(p); (d)-(r)

(4) a-(p); (b)-(r); (c)-(q); (d)-(s)

Ans. (2)

20. A hydrocarbon X that has molar mass 80 gm contains 90% carbon. Find degree of unsaturation in X.

Ans. (3)